Predicting Flight Ticket Prices with Regression Models

DA380A Machine Learning · Lab 1 report · Dataset 5: AirTravel · Mateusz Plizga · 13 October 2026

1. Problem, data and approach

The AirTravel dataset contains 300,261 one-way domestic flights in India between six large cities, with departures between 11 February and 31 March 2022. The target variable is the **ticket price in Indian rupees (INR)**. Predicting it tells a traveller what a trip should cost and shows which factors drive airline pricing. After parsing the price, duration and stop counts (which were stored as text) and removing duplicate and malformed rows, 300,255 flights remained. Ten features were used: airline, class, origin, destination, departure and arrival part of the day, weekday, number of stops, flight duration and days left until departure.

The data was split 80/20, stratified by travel class (240,204 training and 60,051 test flights). All tuning used 5-fold cross-validation on the training set only, and the test set was used once for the final evaluation. Two models were compared: a **linear regression with L2 regularisation (Ridge)**, with one-hot encoded categories and standardised numeric features, and a **Random Forest**, with ordinally encoded categories and no scaling (trees do not need it). One property of the data shapes how the results should be read: prices form two separate groups. The median economy ticket costs about 5,800 INR and the median business ticket about 53,200 INR, so the overall test mean is 20,906 INR with a standard deviation of 22,718 INR.

2. Model performance: MAE, RMSE and R^2

Model (test set, 60,051 flights)	MAE (INR)	RMSE (INR)	R^2	MAE / mean price
Linear (Ridge), default	4,549	6,849	0.9091	21.8 %
Linear (Ridge), tuned	3,277	5,096	0.9497	15.7 %
Random Forest, default	1,120	2,747	0.9854	5.4 %
Random Forest, tuned	1,117	2,741	0.9854	5.3 %

Table 1. Test-set performance before and after hyperparameter tuning. Both models were evaluated on the same test set.

- **MAE** is the typical size of an error in rupees. The tuned Random Forest misses the true price by 1,117 INR on average (5 % of the mean ticket price). The tuned linear model misses it by 3,277 INR (16 %), about three times as much.
- **RMSE** squares errors before averaging, so it is driven by the largest mistakes. For the Random Forest the RMSE is 2.5 times the MAE: most predictions are very close, and a small number of expensive business and last-minute tickets produce large errors. For the linear model the ratio is 1.6, meaning its errors are large across the board rather than concentrated in a few cases.
- **R^2** looks high for both models (0.950 and 0.985), but it is inflated by the two price groups. Simply telling business from economy already explains most of the variance. Table 2 shows the more informative comparison *within* each class, and there the difference between the models is large.

Tuned model	Class	Mean price	MAE (INR)	MAE / mean price	R^2 within class
Linear (Ridge)	Economy	6,589	1,912	29.0 %	0.522
Linear (Ridge)	Business	52,573	6,295	12.0 %	0.598
Random Forest	Economy	6,589	644	9.8 %	0.857
Random Forest	Business	52,573	2,165	4.1 %	0.885

Table 2. Errors of the tuned models on the test set, separately for economy and business tickets.

For the prediction problem this means the following. For an economy ticket the Random Forest is typically 644 INR off (about 10 % of the fare) and explains 86 % of the price variation between economy flights. The linear model is 1,912 INR off (29 %) and explains only 52 %. That is enough to tell economy from business, but too inaccurate to tell a cheap economy flight from an expensive one. Figure 1 shows the same picture: Random Forest predictions lie close to the diagonal, while the linear model spreads widely, especially for business tickets.

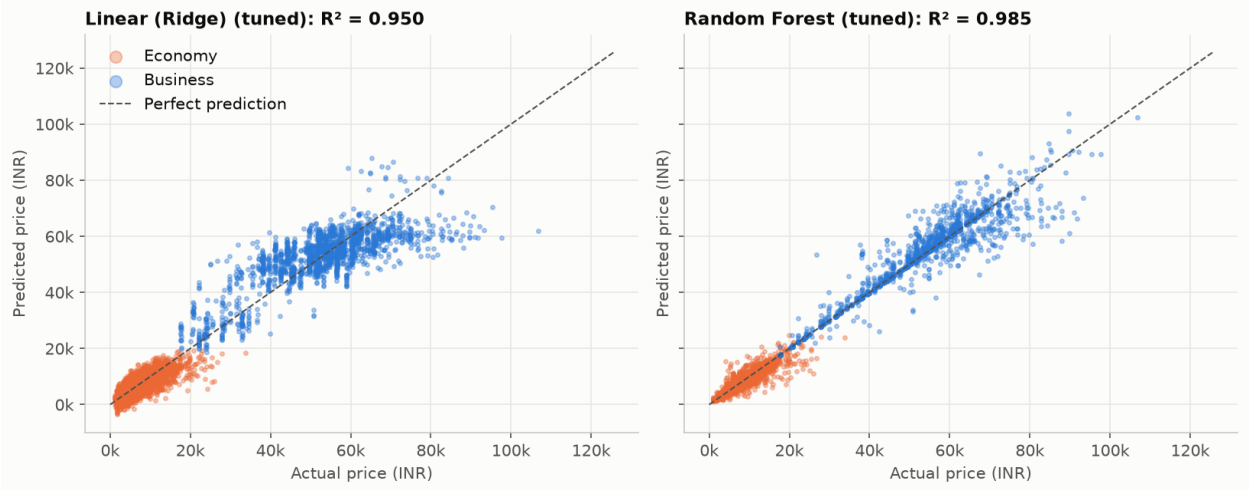


Figure 1. Predicted vs. actual price for 6,000 random test flights (tuned models). The dashed line is a perfect prediction.

Effect of tuning. For the linear model, tuning compared adding pairwise interaction features (yes/no), the regularisation strength alpha (0.01 to 1,000) and a log-transformed target. The best setting (interactions, alpha = 10, untransformed price) reduced the cross-validated RMSE from 6,803 to 5,077 INR (-25.4 %) and raised the test R^2 from 0.909 to 0.950. The interactions account for all of the gain, because they let the model learn class-specific effects. Alpha barely mattered with 240,000 training rows, and the log target increased the absolute errors on expensive tickets. For the Random Forest, a randomized search (25 settings, tuned on a 50,000-row training sample) found that the default was already close to optimal. The best setting (200 trees, max_depth 30, min_samples_leaf 1, all features per split) improved the cross-validated RMSE by only -0.3 %. Every setting that restricted the trees performed worse.

3. Most influential features

Feature influence was measured with **permutation importance**. Each feature was shuffled on 10,000 test flights (5 repetitions) and the resulting drop in R^2 was recorded. The method works for any model and on the original ten features, so both models can be compared directly. For the linear model, the coefficients of the model without interactions were also inspected, because they are readable in rupees.

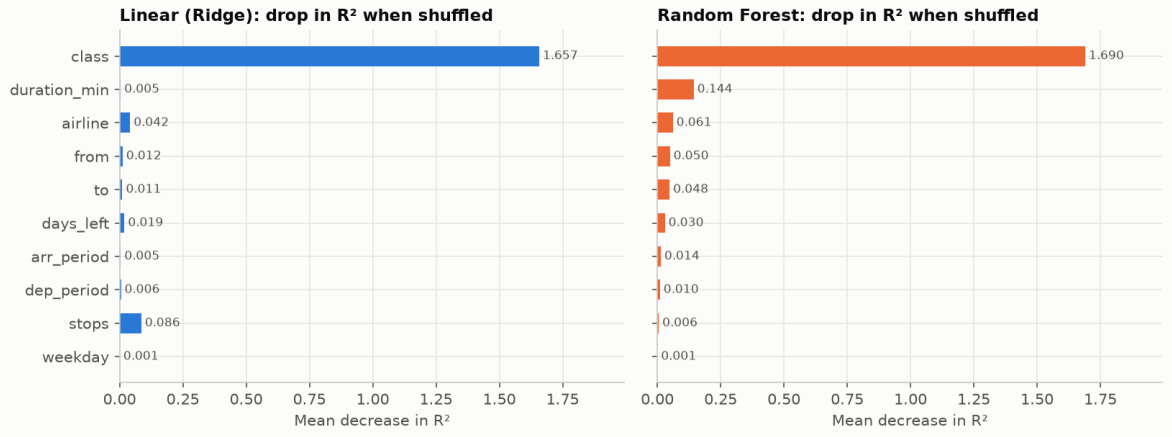


Figure 2. Permutation importance (mean decrease in R^2 when a feature is shuffled) for the tuned models.

- **Travel class dominates both models.** Shuffling it lowers R^2 by about 1.7. The drop exceeds 1 because random class labels make predictions worse than predicting the mean. According to the linear coefficients, business costs about 45,000 INR more than economy when all else is equal.
- **Random Forest:** after class come duration (0.14), airline (0.06), origin and destination (0.05 / 0.05) and days left (0.03). The forest uses duration non-linearly: together with route and airline, it effectively identifies specific itineraries.
- **Linear model:** after class come stops (0.09), airline (0.04) and days left (0.02). Duration matters little (0.005), because its relationship with price is not a straight line. Each extra stop adds about 5,600 INR, and Vistara is the most expensive airline.
- **Booking lead time** has a modest overall importance, but a large effect for late bookings. The linear model estimates that each extra day booked in advance lowers the price by about 128 INR, but Figure 3 shows the real pattern is curved: economy fares roughly triple in the last days before departure. Only the Random Forest captures this curve without manual feature engineering.
- **Weekday** has no influence in either model (importance close to 0), and departure/arrival time of day matter only slightly.

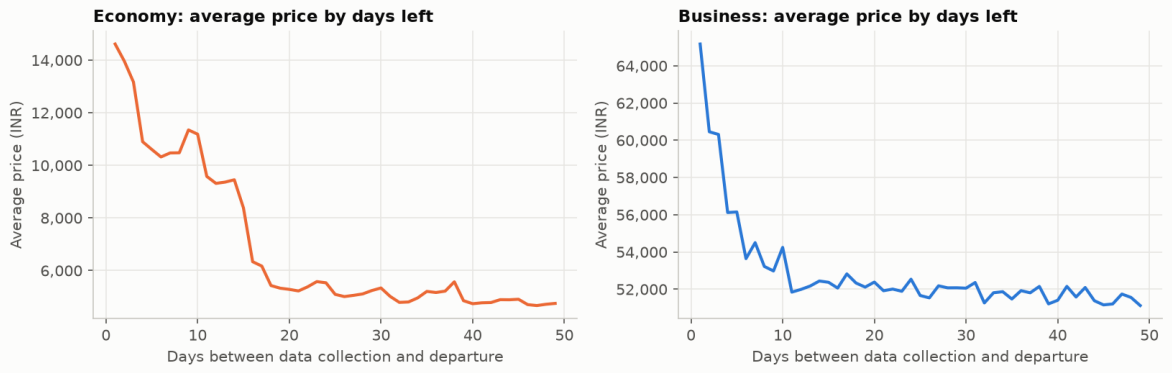


Figure 3. Average price by number of days between data collection and departure, per class (full dataset).

4. Strengths and weaknesses of the two models

	Linear Regression (Ridge)	Random Forest
Strengths	Trains in under a second and gives a small model. The main-effects version is fully transparent: one rupee value per feature, easy to explain to non-experts. Predictions change smoothly and stay sensible outside the training range.	Learns non-linear effects and interactions (e.g. late booking matters more for economy) automatically. Best on every metric and within both classes. Needs no scaling or interaction features and works well with its default settings.
Weaknesses	Assumes effects simply add up, so it cannot follow the curved lead-time pattern or class-specific effects without manual feature engineering. Even with 945 interaction terms its MAE is about 3 times the forest's, and the interaction model is no longer easy to interpret.	Slower to train and predict, and the stored model is large (hundreds of fully grown trees). A black box that needs methods such as permutation importance to explain. Cannot extrapolate beyond the prices and feature values seen in training, and still makes its largest errors on business tickets.

5. Recommendation

The Random Forest is recommended for this dataset. Compared with the best linear model, it lowers the test MAE by 66 % (1,117 vs. 3,277 INR) and the RMSE by 46 % (2,741 vs. 5,096 INR), and explains 98.5 % of the price variance. Most importantly, it stays accurate within each class (R^2 0.86 for economy and 0.88 for business, vs. 0.52 and 0.60 for the linear model). That accuracy is what makes a price prediction useful in practice. The forest also reached this performance with its default settings, so it is robust and needs little tuning. The linear model would only be preferable if a very simple, fully explainable model mattered more than accuracy.

Limitations. The data covers only Indian domestic flights in February-March 2022, and days left is measured from a single data-collection date. The model should therefore be retrained before being used for other periods, markets or booking situations.